

45.5 Models for location, scale and shape

Quantile and expectile regression estimate one feature of the conditional law at a time and never write the law down. The opposite strategy writes it down in full: choose a parametric family with enough parameters to bend as the data require, and let every one of them depend on the covariates through its own structured additive predictor. Rigby and Stasinopoulos (2005) named the result generalized additive models for location, scale and shape.

The gain is a likelihood, and with it penalized estimation, effective degrees of freedom, AIC and Bayesian posteriors. The cost is a parametric assumption about the shape of the conditional distribution, which the data must be allowed to contradict — the subject of [Section 45.6](#).

45.5.1 The model

Definition 45.5.1 — Generalized additive model for location, scale and shape

Let $D(\theta_1, \dots, \theta_K)$ be a family of distributions on the response space with density or mass function $d(y; \theta_1, \dots, \theta_K)$; in this chapter θ_k denotes the k th parameter of that family, not the natural parameter of Definition 34.2.1. A **GAMLSS** asserts that the responses $Y_1, \dots, Y_n$ are independent with

$$Y_i \sim D(\theta_{i1}, \dots, \theta_{iK}), \quad g_k(\theta_{ik}) = \eta_{ik}, \quad k = 1, \dots, K,$$

where each g_k is a strictly increasing **link** mapping the admissible range of θ_k onto $\mathbb{R}$ (the identity for an unrestricted parameter, the logarithm for a positive one, the logit for one in $(0, 1)$), and each η_{ik} is a structured additive predictor (Definition 44.5.1)

$$\eta_{ik} = \beta_{0k} + \sum_{j=1}^{J_k} f_{jk}(v_{ij}), \quad f_{jk}(\cdot) \text{ represented by } \mathbf{Z}_{jk} \boldsymbol{\gamma}_{jk}, \quad (45.5.1)$$

with each $\boldsymbol{\gamma}_{jk}$ carrying a quadratic penalty $\lambda_{jk} \boldsymbol{\gamma}_{jk}^\top \mathbf{P}_{jk} \boldsymbol{\gamma}_{jk}$. The predictors for different parameters may use different covariates and different amounts of structure.

Taking $K = 1$ with D an exponential dispersion family and η linear recovers Definition 34.2.1; with additive η it recovers Theorem 44.6.1. The new content is $K > 1$: the scale, and the shape, become regression problems.

Example. (The heteroscedastic normal model)

The smallest interesting case has $K = 2$, $D = N(\mu, \sigma^2)$, identity link on μ and log link on σ :

$$Y_i \sim N(\mu_i, \sigma_i^2), \quad \mu_i = \eta_{i1}, \quad \log \sigma_i = \eta_{i2}. \quad (45.5.2)$$

The log link keeps $\sigma_i > 0$ whatever the predictor does, as it kept a Poisson mean positive

in Chapter 37. With both predictors linear this is Proposition 45.1.2(b) with normal ε ; with penalized splines it is the fit of Example (Mean and scale together).

45.5.2 Estimation

Theorem 45.5.1 — Penalized likelihood and block-wise Fisher scoring

Write $\ell_i(\boldsymbol{\theta}_i) = \log d(y_i; \theta_{i1}, \dots, \theta_{iK})$ and collect the coefficients of predictor k into $\boldsymbol{\gamma}_k$ with design $\mathbf{Z}_k$, penalty matrix $\mathbf{P}_k$ and smoothing parameter λ_k , so that $\boldsymbol{\eta}_k = \mathbf{Z}_k \boldsymbol{\gamma}_k$. The penalized log-likelihood is

$$\ell_p(\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_K) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta}_i) - \frac{1}{2} \sum_{k=1}^K \lambda_k \boldsymbol{\gamma}_k^\top \mathbf{P}_k \boldsymbol{\gamma}_k. \quad (45.5.3)$$

- (a) **Score.** With $u_{ik} = \partial \ell_i / \partial \eta_{ik}$ and $\mathbf{u}_k = (u_{1k}, \dots, u_{nk})^\top$,

$$\frac{\partial \ell_p}{\partial \boldsymbol{\gamma}_k} = \mathbf{Z}_k^\top \mathbf{u}_k - \lambda_k \mathbf{P}_k \boldsymbol{\gamma}_k. \quad (45.5.4)$$

- (b) **Fisher scoring is penalized weighted least squares.** Let $w_{ik} = -E(\partial^2 \ell_i / \partial \eta_{ik}^2)$ and $\mathbf{W}_k = \text{diag}(w_{1k}, \dots, w_{nk})$. Updating block k with the others held fixed by one Fisher scoring step gives

$$\boldsymbol{\gamma}_k^{\text{new}} = (\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} \mathbf{Z}_k^\top \mathbf{W}_k \mathbf{z}_k, \quad \mathbf{z}_k = \boldsymbol{\eta}_k + \mathbf{W}_k^{-1} \mathbf{u}_k, \quad (45.5.5)$$

the penalized version of the iteratively reweighted least squares of Theorem 34.4.1, with $\mathbf{z}_k$ the working response for parameter k . Cycling over $k = 1, \dots, K$ until nothing changes is backfitting (Theorem 44.2.1) applied to the K blocks.

- (c) **The normal location-scale model is orthogonal.** For Eq. 45.5.2,

$$u_{i1} = \frac{y_i - \mu_i}{\sigma_i^2}, \quad w_{i1} = \frac{1}{\sigma_i^2}, \quad u_{i2} = \frac{(y_i - \mu_i)^2}{\sigma_i^2} - 1, \quad w_{i2} = 2, \quad (45.5.6)$$

and the expected cross-information $-E(\partial^2 \ell_i / \partial \eta_{i1} \partial \eta_{i2})$ is zero. The two blocks are therefore orthogonal: neither Fisher scoring step needs the other's derivatives, the working response for μ is $\mathbf{z}_1 = \mathbf{y}$, and the working response for $\log \sigma$ is $\mathbf{z}_2 = \boldsymbol{\eta}_2 + \frac{1}{2} \mathbf{u}_2$.

Proof.

- (a) By the chain rule $\partial \ell_i / \partial \boldsymbol{\gamma}_k = (\partial \ell_i / \partial \eta_{ik}) \partial \eta_{ik} / \partial \boldsymbol{\gamma}_k$, and $\partial \eta_{ik} / \partial \boldsymbol{\gamma}_k$ is the i th row of $\mathbf{Z}_k$. Summing u_{ik} times that row over i gives $\mathbf{Z}_k^\top \mathbf{u}_k$, and differentiating the quadratic penalty gives $-\lambda_k \mathbf{P}_k \boldsymbol{\gamma}_k$.
- (b) Differentiating Eq. 45.5.4 once more, the contribution of observation i to the second derivative within block k is $\partial^2 \ell_i / \partial \eta_{ik}^2$ times the outer product of the i th row of $\mathbf{Z}_k$ with itself; taking expectations and summing over the independent responses gives $-\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k - \lambda_k \mathbf{P}_k$. A Fisher scoring step therefore reads

$$\boldsymbol{\gamma}_k^{\text{new}} = \boldsymbol{\gamma}_k + (\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} (\mathbf{Z}_k^\top \mathbf{u}_k - \lambda_k \mathbf{P}_k \boldsymbol{\gamma}_k).$$

Adding and subtracting $\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k \boldsymbol{\gamma}_k$ inside the bracket turns the right side into $(\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} \mathbf{Z}_k^\top \mathbf{W}_k (\mathbf{Z}_k \boldsymbol{\gamma}_k + \mathbf{W}_k^{-1} \mathbf{u}_k)$, which is [Eq. 45.5.5](#).

- (c) With $\ell_i = -\log \sigma_i - (y_i - \mu_i)^2 / (2\sigma_i^2) - \frac{1}{2} \log(2\pi)$, $\mu_i = \eta_{i1}$ and $\sigma_i = e^{\eta_{i2}}$, direct differentiation gives $\partial \ell_i / \partial \eta_{i1} = (y_i - \mu_i) / \sigma_i^2$ and

$$\frac{\partial \ell_i}{\partial \eta_{i2}} = \sigma_i \frac{\partial \ell_i}{\partial \sigma_i} = \sigma_i \left\{ -\frac{1}{\sigma_i} + \frac{(y_i - \mu_i)^2}{\sigma_i^3} \right\} = \frac{(y_i - \mu_i)^2}{\sigma_i^2} - 1.$$

Differentiating again, $\partial^2 \ell_i / \partial \eta_{i1}^2 = -1 / \sigma_i^2$, which is already non-random, so $w_{i1} = 1 / \sigma_i^2$; and $\partial^2 \ell_i / \partial \eta_{i2}^2 = -2(y_i - \mu_i)^2 / \sigma_i^2$, whose expectation is -2 , so $w_{i2} = 2$. Finally $\partial^2 \ell_i / \partial \eta_{i1} \partial \eta_{i2} = -2(y_i - \mu_i) / \sigma_i^2$, with expectation zero because $E(Y_i - \mu_i) = 0$. The working responses follow from $\mathbf{z}_k = \boldsymbol{\eta}_k + \mathbf{W}_k^{-1} \mathbf{u}_k$: for $k = 1$, $z_{i1} = \mu_i + \sigma_i^2 (y_i - \mu_i) / \sigma_i^2 = y_i$.

■

Idea. A GAMLSS fit is a stack of weighted smoothers. Each parameter's block is updated by the same penalized weighted least squares step that fitted a generalized additive model, with weights and working responses supplied by the log-likelihood of the chosen family. The rest is bookkeeping, which is why the framework accommodates any family whose derivatives can be written down.

Orthogonality in part (c) is special to this parameterization of the normal family; for most families the cross-information is not zero, and cycling over blocks then converges more slowly or must give way to a joint Newton step. The effective degrees of freedom of a block is the trace $\text{tr}\{(\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} \mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k\}$ that Theorem 27.2.2 attached to ridge regression and Definition 30.1.1 to penalized least squares; the total of these traces goes into AIC (Proposition 29.2.4).

45.5.3 Choosing the family

The distribution is a modelling choice on a par with the link, made from what the response is. Positive with a right tail: gamma, inverse Gaussian, lognormal. Positive with adjustable skewness: the Box–Cox family, whose third parameter is a power estimated rather than assumed, generalizing Definition 22.2.1. Heavy tailed: a t family with the degrees of freedom as a third parameter. Counts with too many zeros: a zero-inflated or hurdle family (Definition 37.5.1). Proportions: a beta family reparameterized into a mean and a precision.

The ancestor is the LMS method of Cole and Green (1992) for growth charts: a Box–Cox power, a median and a coefficient of variation, each a smooth function of age — a three-parameter GAMLSS written a decade before the name, and still how paediatric reference charts are built.

45.5.4 Warnings

Warning (Identifiability). Each predictor in [Eq. 45.5.1](#) contains an intercept and several smooth terms, and a constant moves freely between them. The remedy is that of [Chapter 44](#): constrain every smooth term to sum to zero over the data (Definition 44.1.1), separately in every predictor. Without the constraints the penalized information is sin-

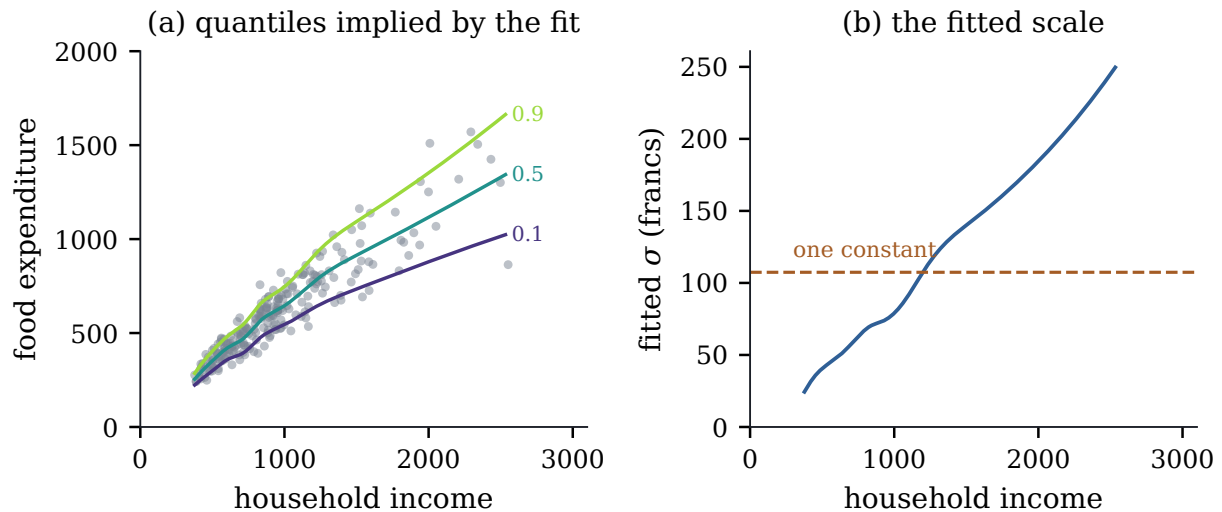


Figure 45.5.1: A normal location-scale GAMLSS for Engel's budgets, both predictors penalized splines in income. (a) The implied 0.1, 0.5 and 0.9 quantile curves. (b) The fitted scale against the homoscedastic constant.

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out
WILD = 0.01                                     # a scale penalty small enough to
run away                                         # thousands of francs

income = data["income"].to_numpy() / 1000.0
food = data["foodexp"].to_numpy() / 1000.0
n = len(food)

lo, hi = income.min(), income.max()
inner = np.quantile(income, np.linspace(0, 1, N_INNER + 2)[1:-1])
KNOTS = np.r_[[lo] * (DEGREE + 1), inner, [hi] * (DEGREE + 1)]

def basis(z):
    """The cubic B-spline basis of Section 43.3, with knots at income quantiles."""
    z = np.clip(np.atleast_1d(z), lo, hi)
    return np.asarray(BSpline.design_matrix(z, KNOTS, DEGREE).todense())

def difference_penalty(q, order=2):
    """K = D'D for the order-th difference matrix D: the penalty of Section 43.4."""
    D = np.diff(np.eye(q), order, axis=0)
    return D.T @ D

def gamlss_normal(Bm, Bs, y, lam_mu, lam_sig, iters=100, tol=1e-9):
    """Penalized Fisher scoring for  $y \sim N(\mu, \sigma^2)$ ,  $\mu = Bm \ b_{\mu}$ ,  $\log \sigma = Bs \ b_{\sigma}$ .

    One block at a time: the mean step is a penalized weighted least squares fit
    with
    weights  $1 / \sigma^2$ , and the scale step is a penalized Fisher scoring step whose
    expected information is the constant 2.
    """
    Km, Ks = difference_penalty(Bm.shape[1]), difference_penalty(Bs.shape[1])
    b_mu = np.linalg.solve(Bm.T @ Bm + lam_mu * Km, Bm.T @ y)
    b_sig = np.linalg.solve(Bs.T @ Bs, Bs.T @ np.full(len(y), np.log(y.std())))
    for _ in range(iters):
        sigma = np.exp(np.clip(Bs @ b_sig, -8, 4))

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    w = 1.0 / sigma ** 2                                # working weights for
the mean
    new_mu = np.linalg.solve(Bm.T @ (w[:, None] * Bm) + lam_mu * Km, Bm.T @ (w *
y))
    u = (y - Bm @ new_mu) ** 2 / sigma ** 2 - 1.0        # score in the log-
sigma direction
    new_sig = b_sig + np.linalg.solve(2 * Bs.T @ Bs + lam_sig * Ks,
Bs.T @ u - lam_sig * Ks @ b_sig)
    step = max(np.max(np.abs(new_mu - b_mu)), np.max(np.abs(new_sig - b_sig)))
    b_mu, b_sig = new_mu, new_sig
    if step < tol:
        break
    return b_mu, b_sig

def neg_log_score(mu, sigma, y):
    """Minus the mean log density: the logarithmic score of Section 45.6."""
    return -np.mean(stats.norm.logpdf(y, mu, sigma))

folds = np.random.default_rng(7).permutation(n) % 10

def cross_validate(design, lam_mu, lam_sig):
    """Ten-fold cross-validated log score of one specification."""
    total = 0.0
    for k in range(10):
        train = folds != k
        Bm, Bs = design(income[train])
        b_mu, b_sig = gamlss_normal(Bm, Bs, food[train], lam_mu, lam_sig)
        Bm, Bs = design(income[~train])
        total += neg_log_score(Bm @ b_mu, np.exp(np.clip(Bs @ b_sig, -8, 4)), food[~
train])
    return total / 10

spline_scale = lambda z: (basis(z), basis(z))
Bm, Bs = spline_scale(income)
b_mu, b_sig = gamlss_normal(Bm, Bs, food, *CHOSEN)
mu, sigma = Bm @ b_mu, np.exp(Bs @ b_sig)
cv_chosen = cross_validate(spline_scale, *CHOSEN)
print(f"smoothing parameters: {CHOSEN[0]:.3g} for the mean, {CHOSEN[1]:.3g} for the
scale")
print(f"cross-validated log score {cv_chosen:.4f}")

```

[Run this code online](#)

```

# barely penalize the scale: the in-sample likelihood climbs, the honest score
collapses
b_mu_w, b_sig_w = gamlss_normal(Bm, Bs, food, CHOSEN[0], WILD)
wild_loglik = np.sum(stats.norm.logpdf(food, Bm @ b_mu_w, np.exp(np.clip(Bs @
b_sig_w, -8, 4))))
loglik = np.sum(stats.norm.logpdf(food, mu, sigma))
wild_cv = cross_validate(spline_scale, CHOSEN[0], WILD)
print(f"log-likelihood: {loglik:.1f} at the chosen penalty, {wild_loglik:.1f} at"
f" lambda_sigma = {WILD:g}")
print(f"cross-validated log score: {cv_chosen:.4f} against {wild_cv:.4f}")

```

[Run this code online](#)

45.5.6 Variations

Bayesian GAMLSS. Every quadratic penalty is a normal prior on the coefficients (Proposition 39.5.1), so Eq. 45.5.3 is a log posterior up to the smoothing-parameter priors, and Section 39.4 transfers whole, with those parameters sampled rather than tuned (Klein, Kneib, Lang and Sohn 2015). **Boosting.** Feeding a GAMLSS log-likelihood to the componentwise boosting of Section 30.4 gives an algorithm that also selects which terms enter which predictor (Mayr, Fenske, Hofner, Kneib and Schmid 2012). **Multivariate responses.** With a multivariate family the dependence parameters become regression problems too; Section 45.7 points at the literature.

45.5.7 Exercises

45.5.7.1 A. Check your understanding

Exercise A1.

In Example (Mean and scale together) the mean block has 12 basis columns and effective degrees of freedom 8.75. What are the two extreme values the effective degrees of freedom of that block could take, and which smoothing parameters produce them?

Solution. As $\lambda_\mu \rightarrow 0$ the trace tends to 12, the number of columns; as $\lambda_\mu \rightarrow \infty$ it tends to the dimension of the null space of the second-difference penalty, namely 2 (constants and straight lines). The fitted value 8.75 is much nearer the flexible end.

45.5.7.2 B. Practice

Exercise B1.

For a gamma response with mean μ and shape ν , density $d(y) = \{\nu/\mu\}^\nu y^{\nu-1} e^{-\nu y/\mu} / \Gamma(\nu)$, with log links on both parameters, compute u_{i1} and w_{i1} of Theorem 45.5.1(b) for the mean block. Is the model orthogonal in the sense of part (c)?

Solution. With $\eta_1 = \log \mu$, $\partial \ell / \partial \mu = \nu(y - \mu) / \mu^2$, so $u_1 = \mu \partial \ell / \partial \mu = \nu(y - \mu) / \mu$. Then $\partial^2 \ell / \partial \eta_1^2 = -\nu y / \mu$, with expectation $-\nu$, so $w_1 = \nu$. The cross derivative $\partial^2 \ell / \partial \eta_1 \partial \eta_2$ involves $\nu(y - \mu) / \mu$ times ν , whose expectation is zero, so the gamma model with these links is also orthogonal — a fact the algorithm exploits, and one that does not extend to three-parameter families.

Exercise B2.

Verify from Eq. 45.5.6 that the working response for the scale block of the normal location-scale model is $z_{i2} = \log \sigma_i + \frac{1}{2} \{(y_i - \mu_i)^2 / \sigma_i^2 - 1\}$, and show that the scale step is therefore an ordinary penalized least squares smooth of the squared standardized residuals, shifted and halved.

45.5.7.3 C. Going deeper

Exercise C1.

Show that the normal family parameterized by (μ, σ) is orthogonal for *any* increasing link on σ , not just the logarithm, and find a parameterization of the normal family in which it is not. What does orthogonality buy in [Theorem 45.5.1\(b\)](#)?